MILITARY SPECIFICATION

STATIC ADJUSTABLE SPEED CONTROL SYSTEM FOR

A. C. INDUCTION MOTOR (ADJUSTABLE FREQUENCY TYPE)

NAVAL SHIPBOARD

1. SCOPE

1.1 This specification covers static adjustable speed control systems by which the speed of an alternating current (a. c.) induction motor is controlled by variation of the input frequency. The equipment covered by this specification utilizes semiconductor devices for power handling.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-C-2212 - Controllers, Alternating-Current, Naval Shipboard.
MIL-C-5015 - Connectors, Electric, "AN" Type.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-F-15160 - Fuses; Instrument, Power and Telephone.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-C-23353 - Connectors, High Reliability, Electrical, Plug and Receptacle, Polarized Shell, for Printed Wiring and Related Applications.
MIL-P-55110 - Printed Wiring Boards.

STANDARDS

MILITARY

MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-195 - Marking of Connections for Electric Assemblies.
MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.

DRAWINGS

BUREAU OF SHIPS

9000-S6202-73907 - Light, Indicator (Switchboard), 2 Lamp, SPF

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N. Y. 10016.)

3. REQUIREMENTS

3.1 Systems. - The adjustable speed control system shall consist of the following individual units:

(a) A static adjustable speed controller designed for continuous operation for furnishing adjustable voltage and frequency to a Service A motor.
(b) A pressure transducer to provide a variable voltage signal to the static controller proportional to a variable input pressure.
(c) An auto-manual control panel to provide for either automatic or manual control of the controller output.

3.2 Performance requirements. -

3.2.1 The static adjustable frequency control system shall be designed to the characteristics specified in 3.2.1.1 through 3.2.1.15.

3.2.1.1 Rating. -

(a) Input - 440 volts (± 10 percent), 60 cycle (± 5 percent), three-phase.
(b) Output - 440 volts (maximum), 60 cycle (maximum) to specified minimum frequency (see 6.1), three-phase, adjustable voltage and frequency (constant volts per cycle over the range of no load to full load).
(c) Controller rating - as required. The maximum output of the controller shall be based on the maximum output current the controller is able to carry as determined by the rating of the semiconductor devices used. For purposes of identification, the identification plate shall indicate the maximum horsepower (h.p.) motor which can be controlled, as well as the maximum output current (see 6.1).

3.2.1.2 Duty. - The equipment shall be designed for satisfactory operation over the range of no load to full load when operating on the input voltage and frequency range specified in 3.2.1.1(a).

3.2.1.3 Ambient temperature. - The equipment shall be designed for operation in an ambient temperature of 50° centigrade (C.) maximum.

3.2.1.4 Insulation. - Insulation shall be class A, B, F, H or N in accordance with MIL-E-917 as selected by the manufacturer.

3.2.1.5 Shock. - The equipment shall be capable of withstanding the high impact shock test as specified for grade A equipment of MIL-S-901 without the use of shock mounts. With rated voltage applied, equipment shall be designed so that it shall not be energized or de-energized because of shock. All devices shall function properly during and after the shock test.

3.2.1.6 Vibration. - The equipment shall be designed to withstand type I vibration tests of MIL-STD-167, without mechanical damage or other malfunctioning.

3.2.1.7 Insulation resistance. - The insulation resistance when conducted in accordance with 4.5.5 shall be not less than 10 megohms.

3.2.1.8 Modes of operation. - The equipment shall be designed for operation in either of the following manners:

(a) Automatic operation of the controller through an input signal from the pressure transducer.
(b) Manual operation of the controller through use of the manual speed adjust rheostat.

3.2.1.9 Dielectric strength. - The equipment shall be capable of withstanding a dielectric strength test (see 4.5.6) of twice the normal voltage of the circuit plus 1,000 volts for 1 minute.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

3. 2. 1. 10 Stability. - The control system shall maintain the output frequency of the controller to within plus or minus one percent of the rated 60 cycle frequency for any steady state input signal to the transducer within the range of no load to full load.

3. 2. 1. 11 Minimum output frequency. - The controller shall be so designed as to limit the output frequency to the minimum value as specified (see 6. 1) in the event that the transducer output voltage goes to zero.

3. 2. 1. 12 Starting. - The controller shall be designed to limit the motor inrush current to between 100 and 150 percent of full load motor amperes or to the maximum of the silicon control rectifier (SCR) limit (if less, but by no means less than 100 percent) and shall provide a smooth frequency increase from zero to 60 cycles within 5 seconds.

3. 2. 1. 13 Airborne noise. - The controller shall be designed to produce a minimum of airborne noise. The airborne noise limits and measurements shall be in accordance with MIL-STD-740 and the following:

(a) Grade C equipment.
(b) Operating conditions - No load to full load at rated voltage.

3. 2. 1. 14 Structureborne noise. - The controller shall be designed to produce a minimum of structure-borne noise and shall be designed for solidly mounting to ship's structure. The structureborne noise limits and measurements shall be in accordance with MIL-STD-740 and the following:

(a) Type 3.
(b) The use of internal sound isolation to meet the requirements of 3. 2. 1. 14 (a) shall be permitted; however, every effort should be made to meet these requirements without their use.

3. 2. 1. 15 Short circuit protection. - Provisions shall be made within the controller for protection of the equipment against faults on the adjustable frequency control system.

3. 3 Detail requirements. -

3. 3. 1 General. - Equipment and components shall be in accordance with MIL-E-917 unless otherwise specified herein.

3. 3. 2 Reliability. - To meet the needs of the Naval service, it is imperative that reliability of operation be considered of prime importance in the design and manufacture of the equipment. The supplier shall employ all reasonable methods possible in the process of manufacture to assure quality and maximum reliability consistent with the state of the art. In order to insure high equipment reliability under all service conditions, avoid marginal applications and provide adequate safety factors, it may be necessary to derate certain parts from their specification values when applying them to equipment circuits. The design shall include all possible features which will result in reliable and stable operation with minimum need for adjustment and alignment, minimum frequency of failure and minimum requirements for maintenance. Maintenance shall be simplified so as to reduce the need for highly skilled maintenance personnel. The final product shall reflect the utmost in simplicity, have maximum reliability and be easy to install and maintain.

3. 3. 3 Interchangeability. - For this equipment, all essential parts including removable units, repair parts and other items, shall be interchangeable physically and electrically with corresponding items or parts without selective matching, drilling, filing, or the use of undue force. Interchangeability of repair parts is necessary to permit both ship and shore maintenance to be performed by the activity using repair parts stocked in the Federal supply system.

3. 3. 4 Life. - Through design and application of parts, the equipment shall be designed for a minimum life of 40, 000 hours.

3. 3. 5 Enclosure. -

3. 3. 5. 1 Degree of enclosure. - The equipment shall be furnished in dripproof protected 45 degree enclosures.

3. 3. 5. 2 Dimensions and weights. - The enclosure shall be as light and compact as possible, while consistent with specified performance and design and shall be capable of passing through a 25 inch hatch. The dimensions shall be as specified (see 6. 1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

3.3.5.3 Materials. - The cabinet structure may be either of steel or aluminum construction.

3.3.5.4 Insulating material. - In order to restrict the use of material which might catch fire or emit fumes when exposed to flame, the quantity of insulating material shall be kept to a minimum and used only where necessary to provide electrical insulation between live parts and to ground.

3.3.5.5 Mounting details. - All equipment enclosures shall be designed for bulkhead mounting. Unless the enclosure is designed to carry the weight of the equipment enclosed, the equipment shall be mounted in such a manner that the weight is transmitted directly to the ship's structure and not borne by the wall of the enclosure.

3.3.5.6 Lifting provisions. - Equipment weighing 75 pounds and over shall be provided with eye-bolts or equivalent means for lifting and handling.

3.3.5.7 Internal temperature. - The design of the enclosure and the placement and selection of parts shall be such that no part will be exposed to a higher operating temperature than the maximum rated temperature for the part.

3.3.5.8 Heating. - Circuit design shall not require the use of heaters to obtain proper operation with varying ambient temperature or the variation in temperature rise within the enclosure due to heating of components under different operating conditions.

3.3.5.9 Accessibility. - All parts which require servicing, repair, replacement or periodic adjustment during the life of the equipment shall be readily accessible through the front of the panel or enclosure; however, provisions for removal of side panels for maintenance may be provided. Wherever practicable, subassemblies shall be utilized in the design of the equipment in order to reduce the time required for service and maintenance. The dimensions of all swing out panels shall be kept to a minimum so as to avoid interference with adjacent equipment.

3.3.6 Component requirements. -

3.3.6.1 Component mounting. - In order to prevent the possibility of displacement of components due to shock and vibration, the use of slotted mounting bolt holes is prohibited. The holes shall be round with a maximum diametrical clearance of 1/16 inch over the mounting bolt diameter. Components previously approved by the bureau or agency concerned shall be mounted in a manner similar to the way they were mounted when subjected to the HI shock and vibration tests during their qualification testing.

3.3.6.2 Component and terminal markings. - All components, terminals and terminal boards shall be marked with the identifying designations shown on the applicable master drawings approved by the bureau or agency concerned. Wherever practicable, the designations shall be in accordance with MIL-STD-195. The identifying designations may be made by metal stamping, engraving, silk screening, stenciling or rubber stamping; however, silk screening, stenciling, or rubber stamping may not be used on terminal boards.

3.3.6.3 Adjustable circuit elements. - The number of adjustable circuits shall be kept to an absolute minimum and limited to those adjustments required by maintenance personnel. When 100 percent adjustable resistors are used, a fixed series resistor shall be employed to protect circuit components from overload due to improper adjustment of the variable resistor. A complete description and instructions for each adjustment shall be included in the manual.

3.3.6.4 Mechanical stability. - All controls of the equipment shall be so designed that when the equipment is properly installed and whether operating or nonoperating, the setting, position, or adjustment of any control shall not change due to normal service conditions.

3.3.6.5 Adjustment locking devices. - Adjustment locking devices, where used, shall be capable of retaining the controls in any given setting within the range of control under shock and vibration. The locking and unlocking action shall be easily and quickly accomplished with the equipment energized and shall not affect the setting of the control. When in the unlocked position, the locking device shall not interfere with the normal operation of the control. Where vernier controls are used, the locking device shall operate on both main and vernier controls.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

3.3.6.6 End stops. - Each control which is not capable of an unlimited degree of motion without damage shall be provided with positive end stops to limit the degree of motion of the control to that within its normal excursion or within its range of calibration.

3.3.6.7 Temperature rise. - Temperature rise of transformers, reactors, and so forth shall not exceed values based upon the limiting hot-spot temperature of 105°C. for class A, 130°C. for class B, 155°C. for class F, 180°C. for class H and 200°C. for class N insulation. Semiconductor temperature rise shall conform to MIL-E-917. The temperature requirements of the referenced parts specifications shall be met, as applicable, for other individual parts.

3.3.6.8 Potted components. - Potting and the use of potted component assemblies shall be limited. Items such as resistors, capacitors and so forth shall not be individually potted. Items to be potted shall be approved by the bureau or agency concerned.

3.3.6.9 Printed wiring. - Printed wiring, where used, shall be limited to control circuits only and shall be designed and constructed in accordance with MIL-STD-275 and MIL-P-55110 and the following:

(a) Interfacial connections shall be made by means of a well formed uninsulated solid copper lead (jumper) extending through a hole and clinched and soldered to the foil on each side of the board so as to contact the conductor or terminal area directly in accordance with MIL-STD-275.
(b) Flared eyelets shall be used for all component terminals in accordance with MIL-STD-275.
(c) Lead connections to the foil shall be either by use of flared shoulder stand off terminals in accordance with MIL-STD-275 or pin and socket type connectors in accordance with MIL-C-23353.
(d) Location of reference designations. - The reference designations shall be located adjacent to each part and shall be marked on the chassis, back of the front panel, partitions or insulator strips. Reference designations shall not be marked on parts which are subject to replacement. The reference designations shall be marked in such a position as to physically locate the parts and yet be readily visible for purposes of maintenance without removal of other parts. The primary intent of this requirement is that removal of a part shall not result in loss of the identification of the physical location of that part. On those printed wiring boards where space does not permit the reference designations to be printed on the board, the designations shall be included as a drawing in the manual.
(e) Printed wiring boards shall be designed to permit replacement of all parts without special tools or techniques.

3.3.7 Cable entrance provisions. - All cable entry, shall be through pin type connectors in accordance with MIL-C-5015 located on the bottom of the cabinet. The controller power input connector shall be female and the connector to the motor shall be male (or vice versa). By this means, in case of controller failure, the two connectors may be removed from the controller and connected together to put the motor directly across the 440 volt line. All connectors shall be identified and so selected that misconnection to any unit is not possible. Each connector shall be provided with its mate for connection to the ship's cable.

3.3.8 Identification and information plates. - The identification and information plates shall be attached to the part of the equipment which will not ordinarily be renewed during its normal service life. These plates shall be located where they can be read at all times without danger to personnel. Identification plates and information plates shall be made of brass, nickel-copper alloy or corrosion resistant steel. The markings shall be either etched, stamped, engraved, metal-photo or anodized aluminum or cast in such a manner as to produce permanent and durable markings to last the anticipated life of the equipment. All etchings, engravings or stampings shall be not less than 0.003 inch deep. The characters on cast plates shall be raised to at least 0.03 inch. All engraved stamped or direct etched markings shall be filled with black paint enamel or lacquer (see 3.3.8.1 and 3.3.8.2).

3.3.8.1 Identification plates. - Identification plates shall be furnished on the outside of the complete unit as well as on major individual components within the enclosure. The minimum data to be marked on the plate shall include the name of the unit, manufacturer's name, manufacturer's serial numbers, Federal stock number, drawing number, contract number, year of manufacture, rating data and a blank space for the Government inspector's official acceptance stamp.

3.3.8.2 Information plates. - The function of each control and each internal adjustment, connector and test point, within the enclosure shall be indicated by means of information plates adjacent to the device.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

3.4 Adjustable frequency controller. -

3.4.1 General. - Controller requirements in addition to those specified in 3.2 and 3.3 shall be as specified in 3.4.2 through 3.4.10.

3.4.2 Type - static. - The static controller is one which does not employ vibrating contacts, carbon piles, electron tubes or mechanically moving parts such as relays (relays for overload protection are permitted) and tachometers. Sliding contact variable resistors and rheostats other than those varied only by manual means will not be permitted. The required performance shall be obtained through circuits using transformers, reactors, magnetic amplifiers, silicon controlled rectifiers, diodes, transistors, rectifiers, resistors, capacitors and other similar solid state devices.

3.4.3 Ventilation. - Two ventilating fans per enclosure shall be provided for forced cooling of the controller. The fans shall be so connected that failure of one fan will not interfere with operation of the other. The ventilation shall be such as to permit operation on one fan only for a minimum of 1 hour without exceeding temperature limits of components. The fan enclosure may be detachable to meet the 25 inch dimensional requirement of 3.3.5.2.

3.4.4 Alarm system. - At least two temperature sensors (one for each fan) shall be installed above the power SCR heat sinks for use in an alarm circuit to indicate high internal air temperature in the cabinet due to cooling fan failure. The circuitry shall be such as to cause a light to turn on in the auto-manual control panel.

3.4.5 Pilot circuit protection. - All control circuit leads between the controller, auto-manual control panel and transducer shall be protected against damage from short circuits. Leads which are in series with current limiting components located within the enclosure are considered to be adequately protected. All other leads to remote devices shall be protected by 10 ampere fuses located within the enclosure of the power source.

3.4.6 Overload protection. - Overload protection shall be provided during both starting and running conditions. The rating of overload relays shall fall within the range of approximately 115 to 125 percent of the motor nameplate current. Overload relays shall be of the hand reset type. Resetting shall be accomplished by means external to the enclosure. When two single-pole overload relays are used with three-phase motors, resetting of the two relays shall be accomplished by a single button.

3.4.7 Undervoltage. - Controllers shall be designed to interrupt the load at 70 percent of rated input voltage.

3.4.8 Overvoltage. - Controllers shall be designed to interrupt the load between 120 and 130 percent of rated input voltage.

3.4.9 Fuses. - Fuses shall be used in series with all power semiconductors and shall be of the type suitable for protection of semiconductor devices. All other fuses shall conform to style F61 of MIL-F-15160.

3.4.10 Overload relays. - Overload relays shall conform to MIL-C-2212.

3.5 Pressure transducer. -

3.5.1 General. - Pressure transducer requirements in addition to those specified in 3.2 and 3.3 shall be as specified in 3.5.2 through 3.5.5.

3.5.2 Operation. - The transducer shall be designed to transform a variable input pressure to a variable output voltage.

3.5.3 Electrical input. - The input to the transducer shall be through a constant voltage source.

3.5.4 Electrical output. - The transducer shall provide a linear variable voltage output with a minimum of hysteresis loss. This voltage shall vary either directly or inversely with the input pressure as specified (see 6.1). The output voltage range shall be such as to provide complete stepless control of the output of the adjustable frequency controller from 60 cycles maximum to the minimum specified frequency (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

3.5.5 Pressure input.- The transducer shall operate over the pressure range specified (see 6.1). The design shall be such as to provide complete stepless control of the output of the adjustable frequency controller from 60 cycles maximum to the minimum specified frequency over the required pressure range.

3.6 Auto-manual control panel.- The auto-manual control panel shall be in accordance with MIL-C-2212 and shall contain the following components:

(a) Auto-manual transfer switch.
(b) Start-stop switch.
(c) Gain adjustment (screw driver slot) to permit adjustment of the frequency reference circuit. By means of this adjustment it shall be possible to raise or lower the output of the controller 15 percent for a set value of transducer input voltage.
(d) Manual adjustment rheostat capable of varying the controller frequency output over its operating range. An instruction plate shall indicate the direction of frequency increase or decrease and shall be calibrated in percent of full output frequency.
(e) A red alarm light for each fan shall be mounted on the front of the panel and shall be lighted in the event of excessive air temperature in the cabinet due to a controller ventilation fan failure. In the case of two or more controller cabinets, one alarm light shall be provided for each cabinet. An identification plate shall be provided for each light.
(f) Indicating lights.- A.c. indicating lights in accordance with Drawing 9000-s6202-73907 shall be provided on the front of the cabinet with the following colors:
(1) Power available to the controller - White.
(2) Normal operation - Green.

3.7 Onboard repair parts.- Onboard repair parts shall be as follows:

(a) Transducer - One set of diaphragms, O-rings and seals for each set installed per ship.
(b) Static controller and auto-manual control panel - As specified in table I.

Table I - Quantities of static controller and auto-manual control panel onboard repair parts to be furnished per ship.

Parts required	Identical parts, or sets 1/ of parts, installed per ship							
	1	2	3	4	5-8	9-20	21-50	51-100
Capacitors, each size and type	1	1	1	1	2	3	4	5
Fuses, sets	1	1	1	1	2	3	4	5
Rectifiers, each size and type	1	2	3	3	4	6	7	8
Resistors, each size and type	1	1	1	1	2	3	4	3
Rheostats, each size and type	1	1	1	1	2	3	4	5
Saturable reactors only if the reactors contain magnet wire of less than 1000 circular mils cross-sectional area complete set (or four, whichever is less)	1	1	1	1	1	2	3	4
Transistors, each size and type	1	2	3	3	4	6	7	8
Voltage regulator diodes	1	2	3	3	4	6	7	8
Voltage reference diodes	1	2	3	3	4	6	7	8
Potted components as approved by the bureau or agency concerned	1	2	3	3	4	6	7	8
Printed wiring boards 2/	1	1	1	1	1	-	-	-

1/ A set is all parts of the same size and type installed in one controller.
2/ A board shall be defined as the laminated wiring board plus all components normally mounted thereon.

3.7.1 Matched parts.- Where matching of parts is required in the equipment, the matched set of parts shall be considered one part and furnished as such if required as a repair part.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

3.8 Drawings.- Drawings for the static controller, pressure transducer and auto-manual control panel shall be class A in accordance with MIL-D-963 and shall contain the following data:

(a) Descriptive data for the equipment.- .
- (1) Type of equipment.
  - a. Static adjustable frequency, adjustable speed controller, transducer or auto-manual control panel.
- (2) Manufacturer's identification.
  - a. Type, class, model number.
- (3) Reference specification.
  - a. A statement as follows: The equipment shown hereon conforms to Specification MIL-S-24158 with the following exceptions: (State "none" or list all exceptions as applicable.)
- (4) Equipment classification (as applicable).
  - a. Type of equipment.
  - b. Enclosure.
  - c. Shock classification.
  - d. Ambient temperature.
  - e. Electrical ratings, voltage, current, phase, frequency, h.p., and so forth.
  - f. Mounting.
  - g. Ventilation.
  - h. Weight of complete assembly.
  - i. Insulation class.
  - j. Special features.
- (5) Test reports
  - a. The file number and date of the noise, shock and vibration tests and of the approval letters shall be referenced on the drawing.
  - b. The periodic test report number or numbers and the file number and date of approval letter shall be referenced on the drawing.

(b) Parts design data.-
- (1) Transformers, reactors and magnetic amplifiers.
  - a. Core material.
  - b. Winding data including the number of turns, tape, wire size and type, terminal identification, direct current (d.c.) resistance between terminals at 25°C., nominal voltage between terminals, volt-amp ratings for transformers, and current ratios for current transformers.
  - c. Part designation as assigned in the elementary schematic diagram and part number used in the list of material.
  - d. Complete material details including applicable specifications for materials used. Details of core boxes, layer and coil insulation, ground insulation, method of impregnation and treatment, potting methods and materials, terminal boards, grommets and leads shall be included.
- (2) Resistors.
  Style, resistance, voltage rating, wattage, type designation or part number, type of mounting, type of terminals, type of adjusting band and any other special data which may be necessary for ordering exact duplicates from the resistor manufacturer.
- (3) Rheostats.
  Nominal diameter, resistance, wattage, voltage rating, current rating and part number.
- (4) Capacitors.
  Style, capacitance, voltage rating, type designation or part number. Also indicate the maximum a.c. (peak and root mean square (r.m.s.)) and d.c. (average) voltages which will be applied to the capacitor in the application and the condition at which these maximum voltages occur (transient load conditions, 50 percent rated current, and so forth).
- (5) Semiconductor devices.
  Each device shall be identified in the list of material by manufacturer's part or drawing number to permit direct ordering (Joint Electron Devices Engineering Council (JEDEC) number is adequate). Also provide mounting details and application (not device capability) data as follows:
  - a. Rectifiers using silicone diodes.
    1. Circuit symbols and piece number.
    2. Maximum reverse voltage peak and r.m.s. (complete rectifier circuit).
    3. Maximum load current d.c. amperes (complete circuit).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

4. Maximum diode case temperature at design ambient-use for diodes rated against core temperature.
5. Maximum ambient temperature directly adjacent to diodes-use for diodes rated against ambient temperature - no cooling plates.
6. Cooling plate size, thickness and material where so mounted.
7. Air velocity if forced cooled (linear feet per minute).
8. Total diodes in circuit.
9. Description of insulation hardware if insulated from a cooling plate.

b. Voltage regulator diodes.
1. Circuit symbol(s) and piece number.
2. Range of actual voltage levels which circuit will accept (with or without circuit adjustment from regulator diodes) and still give required equipment performance_____volts to_____volts.
3. Regulation (including temperature error) permitted in an installed regulator diode under all conditions of operation (assume equipment is initially adjusted for optimum performance ± _____mv).
4. Maximum level of steady state average reverse current through regulator diode (steady state shall be considered any time in excess of 30 seconds):_____ma.
5. Maximum transient reverse current under any operating condition_____ma for_____sec duration and_____frequency of occurrence.
6. Maximum ambient temperature at diode (no cooling plate).
7. Maximum case temperature of diode (complete for diodes rated against case temperature).
8. Cooling plate size, thickness and material where used.
9. Air velocity if forced cooled (linear feet per minute).
10. Description of insulation hardware if insulated from a cooling plate.

c. Voltage reference diodes (see regulator diodes also).
1. Circuit symbol(s) and piece number.
2. Range of reference voltage levels which circuit will accept (with or without circuit adjustment) from the voltage reference diode(s):_____volts to_____volts.
3. Error permitted in an installed voltage reference diode under all conditions of operation: ±_____millivolts.
4. Circuit design current through voltage reference.
   (a) Nominal or average value:_____ma.
   (b) Regulation about nominal value: ±_____ma.
5. Maximum ambient temperature (include internal heat rise above outside design ambient):_____a.

d. Transistors.
1. Circuit symbol(s) and piece number.
2. Maximum collector to emitter voltage_____volts.
3. Maximum collector to base voltage_____volts.
4. Maximum collector current_____ma.
5. Maximum base current_____ma.
6. Maximum collector power dissipation _____mw.
7. Maximum ambient temperature (include internal heat rise at transistor at 50°C.) design ambient_____°C.
8. Maximum core temperature of transistor at 50°C. equipment design ambient and any condition of required equipment operation_____°C. (Where rating is based on case temperature.)
9. Air velocity if forced cooled.
10. Cooling plate size, thickness and material where so mounted.
11. Description of insulating hardware if insulated from a cooling plate.

e. Selenium surge suppressors.
1. Circuit symbol(s) and piece number.
2. Supply voltage, steady state r.m.s. (nominal and maximum).
3. Peak discharge current.
4. Peak and average power dissipation.
5. Maximum cell temperature.
6. Peak surge voltage with suppressor(s) applied.
7. Circuit arrangement and terminal marking.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

f. Overload heater rating.
g. Printed wiring boards.
In addition to the drawing requirements of MIL-STD-275, the following drawing shall be provided for inclusion in the manual:
1. An enlargement of the foil pattern shall be shown with each component identified and shown in phantom.

(c) Schematic diagram.-
(1) A schematic diagram of the complete system to clearly represent the operation of the equipment. The following features shall be incorporated:
a. Each part, such as resistors, capacitors, and so forth shall be identified by symbol (R-resistor, T-transformer, L-reactor, and so forth). Each symbol shall be used only one time in the complete power supply.
b. In addition to the part designation, the numerical value of all resistances, capacitances and inductances shall be noted adjacent to the part.
c. Identification of all terminals shall be shown.
d. A brief description of theory of operation of the system to enable the approval authority to understand the circuitry.
e. Simplicity in reading and understanding the operation of the system shall be stressed throughout.
f. The main power lines should be drawn heavier than the control lines for ease in tracing power circuit.

(d) Wiring diagram.-
(1) A wiring diagram for each of the major subassemblies and interconnection of the subassemblies. The following features shall be incorporated in the drawing:
a. All parts shall be shown in correct relation to physical location within a particular enclosure.
b. Terminals of all assemblies and subassemblies shall be clearly shown and marked.
c. Designation of parts shall conform to those assigned in the schematic diagram. Also, the piece numbers corresponding to those shown in the list of material shall be indicated.
d. Wire size, type, applicable specifications and type of terminals shall be shown in tabular form.
e. Lead identification.
f. Continuous lines to represent leads between components need not be drawn if the lead identification used clearly indicates the termination point of each lead.

(e) Construction details.-
(1) Two representative views of each enclosure showing the following:
a. Overall dimensions of the enclosure.
b. Drilling and mounting dimensions.
c. Outline of all parts, subassemblies and assemblies contained in each enclosure identified with item or piece number corresponding to item or piece number contained in the list of material.
d. The parts need not be drawn in detail nor to scale but should represent general configuration of the part and show its general location in the enclosure.
e. Outline drawing of parts or components furnished separately (not mounted in an enclosure) to show critical dimensions, including mounting dimensions.

3.8.1 Installation drawings.- Data to be shown on installation drawings shall be as follows:

(a) Dimensional outline of each unit, showing overall and mounting dimensions in sufficient detail to establish the limits of space in all directions required for installation, operation and servicing, exclusive of space required for personnel. The amount of clearance required to permit opening hinged panels and doors, for withdrawal of parts and subassemblies, for cable access and for ventilation shall be included.
(b) Weight of each enclosure and separately furnished assembly.
(c) Center of gravity of each unit.
(d) A block diagram of each unit, showing interconnection between the units. Terminal identification, as well as maximum current rating of each cable, should be shown.

3.9 Manuals.- The dimensions of the manual shall be 11-5/8 inches high and 8-3/4 inches wide. Manuals shall conform to type I of MIL-M-15071 and shall contain the following information in the order listed:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

(a) Front material.
  (1) Cover.
  (2) Approval and procurement record page.
  (3) List of effective pages.
  (4) Table of contents.
  (5) List of figures.
  (6) List of tables.
  (7) List of drawings.

(b) Chapter 1 - General information.
  (1) Section 1 - General data.
    a. Table of units furnished with dimensions and weights indicated.
    b. Power supply requirements (voltage, frequency, current, phase permitted voltage and frequency variation, and so forth).
    c. Photographs of equipment with doors open and closed.
  (2) Section 2 - Introduction.
    a. Brief description of equipment furnished its basic parameters and overall system operation.
    b. Block diagram of complete system, indicating all pertinent wave shapes between major controller components.

(c) Chapter 2 - Detailed description of operation.
  (1) Section 1 - Detailed description of operation of adjustable frequency controller, transducer and auto-manual control panel with schematic diagrams of various assemblies of the controller where essential to the understanding of its operation.

(d) Chapter 3 - Installation.
  (1) Section 1 - Installation instruction.
  (2) Section 2 - Recommended procedure and adjustments for initial starting.

(e) Chapter 4 - Operating instruction.
  (1) Section 1 - Startup.
  (2) Section 2 - Automatic operation.
  (3) Section 3 - Manual operation.
  (4) Section 4 - Shutdown.
  (5) Section 5 - Emergency operation (power failure, partial failure of equipment, and so forth).

(f) Chapter 5 - Maintenance.
  (1) Section 1 - Safety precautions.
  (2) Section 2 - Preventive maintenance.
    Routine inspection and care of equipment.
  (3) Section 3 - Corrective maintenance.
    Wave shapes - controller output at 1/4, 1/2 and full load and between major controller components. Trouble shooting charts.
    Tables showing points of measurements of voltage, current, resistance and so forth.
    Adjustments after repair.
  (4) Section 4 - Repair of printed wiring modules.
  (5) Description of test equipment necessary for maintenance and trouble shooting.

(g) Chapter 7 - Parts identification.
  (1) Section 1 - Introduction.
  (2) Section 2 - Parts tabulation.

(h) Chapter 8 - Drawings - reduced size prints of approved adjustable frequency controller, transducer and auto-manual control panel, installation drawing and printed module foil patterns.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

4.1.1 The manufacturer's in-process inspection shall include the applicable items of inspection contained in Appendix I herein, selected in accordance with the design, manufacturing methods and sequence of operations. In-process inspection procedures, measuring gauges, instruments and written records shall be acceptable to the Government.

4.2 Routine tests. - All equipment offered for delivery shall be subjected to the routine tests specified in table II. Nonconforming equipment shall be individually rejected. Routine tests which require the equipment to be operated as a system shall be conducted, utilizing a new set of controller, transducer and auto-manual control panel for each test series. This restriction shall be for the purpose of testing each component as required and shall not be construed as a requirement for matched sets. Each unit shall be capable of operating with any two others.

4.3 Periodic tests. - The first adjustable frequency control system of each design and size offered for delivery shall be subjected to the periodic tests specified in table II. Thereafter one periodic test shall be required on identical equipment every 12 months. However, shock, weight measurement and vibration tests are required only once on any specific design of equipment. If the control system subjected to the periodic tests fails in any test, the control system tested and those of the same design being offered for delivery shall be rejected. After the design has been modified and the revisions incorporated into the system, the system may be resubmitted for tests. A periodic test shall be required after any change in design which affects the performance characteristics. If routine test data reveals variations beyond a normal manufacturing tolerance, the Government may require that any or all of the periodic test be made on a particular equipment to demonstrate that it conforms to this specification.

Table II - Routine and periodic tests.

Description of test	Applicable test paragraph		Tests to be conducted while equipment is operating together as a complete system	Tests To be conducted on controller alone	Tests to be conducted on transducer alone	Tests to be conducted on control panel alone
	Routine test	Periodic test				
Creepage and clearance	4.5.1	4.5.1	------	X	X	X
Enclosure	4.5.2	4.5.2	------	X	X	X
Weight	-----	4.5.3	------	X	X	X
Vibration	-----	4.5.4	------	X	X	X
Insulation resistance	4.5.5	4.5.5	------	X	X	X
Dielectric strength	4.5.6	4.5.6	------	X	X	X
Shock	-----	4.5.7	------	X	X	X
Voltage variation	-----	4.5.8	X	------	------	------
Temperature rise	-----	4.5.9 and 4.5.18	X	X	------	------
Noise	4.5.10.2	4.5.10.1	------	X	------	------
Stability	4.5.11	4.5.11	X	------	------	------
Minimum output frequency	-----	4.5.12	X	------	------	------
Controller output	4.5.13	4.5.13	------	X	------	------
Ventilation	-----	4.5.14	------	X	------	------
Manual adjustment	-----	4.5.15	X	------	------	------
Gain adjustment	-----	4.5.16	X	------	------	------
System test	4.5.17.2	4.5.17.1 and 4.5.17.1.1	X	------	------	------

4.4 Test reports and approval. -

4.4.1 Test results shall be recorded by the supplier on test forms prepared by the supplier and approved by the Government. The test data shall be bound into 9 by 11-1/2 inch binders. Each test report shall be identified by the number of this specification and by sufficient information to identify the equipment. No special format will be required; however a table of contents shall be included at the beginning of the report and list of each test required by table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

4.4.2 Each test shall be prefaced by the name of the test and applicable test paragraph as indicated in table II. The data submitted shall be copies of the actual data taken on the test floor and not retyped data. The forms used shall allow sufficient columns to make instrument corrections and necessary calculations. When oscillograms are required, each trace shall be identified as to the circuit measurement involved, and its calibration shall be indicated to permit ready use without reference to associated documents. Periodic noise tests shall be detailed reports in accordance with MIL-STD-740.

4.4.3 All test data shall be approved by the Government inspector prior to the shipment of the equipment. Upon approval, one file copy of all periodic test data shall be forwarded to the bureau or agency concerned.

4.5 Tests. -

4.5.1 Creepage and clearance. - Creepage and clearance distances between current carrying parts in the equipment shall be measured to determine compliance with MIL-E-917.

4.5.2 Enclosure. - Acceptability of the enclosure shall be determined by visual examination or measurements to determine compliance with 3.3.5.

4.5.3 Weight. - The weight of the static controller, pressure transducer and auto-manual control panel shall be measured and recorded.

4.5.4 Vibration. - Equipment shall be subjected to the type I vibration test of MIL-STD-167. The equipment shall be mounted in normal operating position. Operation shall be checked at each frequency of vibration.

4.5.4.1 Definition of failure. -

(a) Breakage or appreciable distortion of any parts, including mounting bolts.
(b) Electrical malfunctioning. All design characteristics shall continue to be met. Protective features shall not malfunction or become inoperative.
(c) Low insulation resistance. After vibration test, the insulation resistance shall conform to 4.5.5.
(d) Failure to pass examination. The equipment shall be disassembled following checks (a) through (c) and examined thoroughly for damage. The extent of disassembly need be only to the point where the condition of the equipment can be easily observed. Any observed effects of the vibration should be recorded.

4.5.5 Insulation resistance (cold). - This test shall be made before the dielectric tests. Prior to the application of the test voltage, all circuits shall be thoroughly discharged by shorting them to the frame. The test voltage shall be applied between all electrically isolated circuits and between each circuit and ground. Circuit diagrams should be carefully studied prior to conducting this test to ascertain that circuits which may be isolated by transformer are not inadvertently neglected. When testing between the circuits and ground, all circuits of the control system may be tied together so that only one test voltage need be applied to each component providing that the insulation resistance when tested in this manner shall be as specified in 3.2.17. The measurement shall be made with an insulation resistance indicating meter with a full scale reading of 100 or 200 megohms, an open circuit voltage of 500 volts and a voltage of at least 45 volts across a resistance of 1 megohm. The temperature and relative humidity shall be measured and recorded. Insulation resistance measurements shall be corrected to 25°C. Corrections shall be made on the basis of insulation resistance doubling for each 15°C. decrease in temperature. No correction shall be made for humidity.

4.5.6 Dielectric tests. -

4.5.6.1 The dielectric test shall be made after all other tests have been completed. The dielectric test shall be made on the completely assembled units and not upon individual parts within a unit. An exception is made in the case of repair parts which require dielectric tests.

4.5.6.2 Points of application. - The test voltages of a magnitude as specified in 3.2.1.9 shall be successively applied between each electric circuit and all other electric circuits and metal parts. No component, however, in an assembled unit shall be subjected to a dielectric test voltage higher than that test voltage required for that component by the individual component specification. When the unit test voltage is higher than that required for an individual component, that component shall be disconnected from the circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

4.5.7 Shock. - Shock tests shall be conducted in accordance with MIL-S-901 for grade A, class I equipment as specified in 4.5.7.1 and 4.5.7.2.

4.5.7.1 Method of mounting. - The equipment shall be energized during the test. All equipment or parts shall be mounted on the shock machine in a manner simulating the actual installation on board ship. The weight designation of the shock test shall be as required by the combined weight of the equipment being tested. No adjustments in input or output shall be made between blows.

4.5.7.2 Definition of failure. -

(a) Breakage or appreciable distortion of any parts, including mounting bolts.
(b) Electrical malfunctioning. The conditions of 3.2.1.5 shall apply. Protective features shall not malfunction or become inoperative.
(c) Low insulation resistance. After shock tests, the insulation resistance shall conform to 4.5.5.
(d) Low dielectric strength. After shock tests, the dielectric test shall be conducted at a voltage equal to 65 percent of the voltage required by 3.2.1.9.
(e) Failure to pass examination. The equipment shall be disassembled following checks (a) through (d) and examined thoroughly for damage. The extent of disassembly need be only to the point where the condition of the equipment can be easily observed. Any observed effect of the shock should be recorded.

4.5.7.3 Disposal of shock tested equipment. -

(1) Equipment which has been subjected to the high impact shock test and has failed to perform any of the principal functions specified herein will not be acceptable, either in whole or any of the parts.
(2) Equipment which has been subjected to the high impact shock test and has successfully performed all the principal functions specified herein will be acceptable under a contract or order after correction of any minor damage.

4.5.7.4 The place at which the shock test shall be conducted. - Shock tests may be performed at the contractor's plant, commercial laboratory, or Government laboratory which is suitably equipped to perform these tests. In either case, the tests are to be conducted at the expense of the contractor. Where shock tests are conducted at a Government laboratory, copies of all master drawings shall accompany the equipment.

4.5.7.5 Shock extension. - The extension of shock test from one unit of equipment to another shall be based on comparision of size, weight, and component distribution. The manufacturer shall submit a letter of request together with the drawings of each unit. Upon receipt of approval of the bureau or agency concerned, the manufacturer shall note on the master drawing the test number and date of original shock test, the Governments approval letter, serial number and date and a statement of shock extension including the approval letter of the bureau or agency concerned.

4.5.8 Voltage variation. - The controller shall be tested to determine conformance with 3.2.1.1(a), 3.4.7 and 3.4.8. It shall be determined that the controller will not malfunction due to the input voltage varying between 90 and 110 percent of rated value from the range of no load to full load. Reading shall be taken at 1/4, 1/2 and full load. The voltage shall then be decreased to 70 percent to determine that the controller de-energizes the load at 70 percent of rated input voltage. The voltage shall be increased to 120 percent to determine that the controller de-energizes the load between 120 and 130 percent of rated input voltage. Failure to meet these requirements shall be cause for rejection.

4.5.9 Temperature rise. - A temperature rise test shall be performed with the controller operating at a loading which produces maximum heating in the unit. The test methods to be employed and the precautions to be observed shall be as specified in 4.5.9.1 through 4.5.9.4.

4.5.9.1 Assembly and mounting of equipment. - Temperature rise tests shall be made only with the equipment completely assembled and mounted in the manner for which it is designed. Barriers shall be placed adjacent to the enclosure, at the manufacturer's recommended minimum clearance for adequate ventilation, to simulate shipboard space restrictions. Temperature rise tests shall be made on all main semiconductor studs, coil windings including power transformers and reactors) and rectifier cells except for those rectifier cells of control circuits which clearly do not contribute to the overall temperature rise within

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

the equipment enclosure and except for those which are tested and given separate qualification approval under its individual specification. For those components which are given exception based on separate qualification approval, it shall be determined that its adjacent ambient within the enclosure does not exceed the rated ambient of the component and that the application rating does not exceed the manufacturer's approved rating at the applied temperature ambient.

4.5.9.2 Method of temperature measurements. - The method of temperature measurement shall be either the thermometer method or embedded detector method in accordance with MIL-E-917. The temperature measuring elements shall be applied to the hottest accessible part of the component whose temperature is to be measured. The number of measuring elements shall be liberal and so placed as to ascertain the highest temperatures. Surface mounted thermometers or thermocouples shall have their bulbs covered by felt pads or by oil putty. The bulbs shall be placed in such a position that they make the maximum practicable line contact with the part whose temperature is to be measured, and they shall be so firmly supported that this degree of contact will not be altered by gravity and normal vibration. When the thermometer method is used the true temperature rise shall be determined by adding the value of temperature gradient to the indicated temperature rise. The temperature gradient shall be predetermined prior to assembly of equipment by taking measurements of maximum temperature of the components under rated application load by both the resistance method and the thermometer method. The difference in measurements between the thermometer method and the resistance method shall be the value of temperature gradient. When the embedded detector method is used the true temperature rise shall be considered the indicated temperature rise of the detector element giving the maximum value. The temperature rise of all encapsulated or potted components shall be determined by embedded thermocouples in the samples to be used for this test. The case or cell temperature required for rectifiers and other semiconductor devices relying upon a cooling fin or other surface to dissipate heat from the case to the ambient shall be measured by placing a thermocouple in direct contact with the case (on center cell near through bolt or stud of a selenium rectifier stack). Position on the case shall be on the high temperature side of the case. In no instance shall the thermocouple be placed on the cooling fin or other heat dissipating surface. (The selenium cell or plate is an exception since the rectifying junction is distributed across the cell.)

4.5.9.3 Duration of test. - The temperature rise test shall be continued until the temperature rise of components that can be observed during test have attained a steady final value.

4.5.9.4 Ambient temperature. - The equipment shall be tested at any convenient room temperature above 10°C., but whatever the value of this ambient temperature, the maximum permissible temperature (hot spot temperature) as specified in MIL-E-917 for the class insulation used shall not be exceeded when converted to the rated ambient of the equipment. The ambient temperature of semiconductor devices shall be measured by method 1 of MIL-E-917 within 0.5 inch of the device on the low temperature side (side from which air starts passage over device).

4.5.10 Noise. -

4.5.10.1 By instrument (for periodic test). - Airborne and structureborne noise tests on the controller only shall be performed in accordance with MIL-STD-740. The tests shall be performed at that load point at which maximum noise occurs in the controller.

4.5.10.2 By observation (for routine tests). - Observation shall be made of equipment noise during the progress of testing to determine that the equipment is free from any degree of noise comparably greater than that inherent in the given type and size of unit. If any objectionable noise is detected, the Government may require a test by instrument as specified in 4.5.10.1 to ascertain that the unit meets the requirements of 3.2.1.13 and 3.2.1.14.

4.5.11 Stability. - During the tests specified in 4.5.17.1 and 4.5.17.2, the output voltage and frequency shall be recorded at the start, middle and end of each series of steady state tests to determine compliance with 3.2.1.10.

4.5.12 Minimum output frequency. - With zero input signal to the controller, the output frequency shall be recorded to determine compliance with 3.2.1.11.

4.5.13 Controller output test. - The maximum and minimum value of output frequency shall be recorded at rated input conditions to determine compliance with 3.2.1.1(b).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

4.5.14 Ventilation systems test. - With the controller operating at full load, one fan shall be disconnected, and the time in which the controller reaches its limiting temperatures (or levels off) shall be recorded (as well as the temperature readings) to determine compliance with 3.4.3.

4.5.15 Manual rheostat adjustment. - The manual rheostat shall be varied from maximum to minimum to demonstrate compliance with 3.6(d).

4.5.16 Gain adjustment. - The gain adjustment shall be adjusted from maximum to minimum for both maximum and minimum controller input signal to determine compliance with 3.6(c).

4.5.17 System tests. - The following tests shall be performed with the static controller, transducer and auto-manual control panel actually controlling a motor. The motor used for these tests shall be preferably a Service A motor of appropriate horsepower and current rating; however, a commercial motor may be used provided it has been previously tested at rated horsepower output and operates satisfactorily within its insulation class. Full load heat run data on the motor operating at rated voltage and frequency shall be included in final test report. A means shall be provided to continuously vary the input pressure to the transducer. Dynamometer loading of the motor is acceptable and shall be sequenced with the transducer output signal changes so that the load changes from full load at maximum output signal to no load (approximate) at minimum output signal.

4.5.17.1 For periodic tests. -

Hours	Loading
50	Rated horsepower and frequency.
10	Minimum horsepower and frequency.
40	Cycling 30 cycles per hour. One cycle shall involve a change of input pressure to the transducer from minimum to maximum and back to minimum.

4.5.17.1.1 There shall be a series of 10 starts and stops during each of the test runs. An oscillogram of motor current, voltage, frequency and their wave shapes shall be taken during the first and last start in each of the test runs.

4.5.17.2 For routine tests. -

Hours	Loading
8	Rated horsepower and frequency.
8	Cycling 30 cycles per hour as indicated in 4.5.17.1.

4.5.17.3 During each of the above tests, the data as required in 4.5.9.1 and the following shall be measured and recorded:

(a) Motor stator and rotor winding temperature data and input watts (periodic tests only).
(b) During the first five cycles of the 1st, 20th and 40th hours of the cycling test of 4.5.17.1, the transducer output voltage shall be recorded on an oscillogram. The corresponding value of input pressure shall also be noted on the oscillogram.
(c) Controller input current, voltage and frequency and output voltage, current and frequency shall be recorded every hour for at least one full cycle.

4.5.17.4 The tests specified in 4.5.17.1 and 4.5.17.2 shall be performed without failure. Any failure during these tests shall invalidate the entire test. After repair of the equipment, the tests shall be re-run. If continuous, uninterrupted testing is not practical, the minimum testing period shall be 8 hours at any one time.

4.5.18 Component ratings. - The rating of all components, such as resistors, rectifiers, capacitors, transformers, instruments, rheostats, and so forth shall be as shown on the approved master drawing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

4.5.19 Repair parts. - Repair parts shall be subjected to the following tests as applicable:

(a) Insulation resistance (see 4.5.5).
(b) Dielectric (see 4.5.6).
(c) Creepage and clearance (see 4.5.1).
(d) Materials.
(e) Resistance.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Basic equipment. -

5.1.1.1 Preservation and packaging. - Preservation and packaging which may be the supplier's commercial practice shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at lowest rate and will afford protection against physical or mechanical damage during shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. - Onboard repair parts shall be preserved and packaged level A; packed level C and marked levels A and C, respectively, in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 Level A. -

5.2.1.1.1.1 Basic equipment. - The basic equipment shall be preserved and packaged level A in accordance with MIL-P-16298. When the size and mounting arrangement of the basic equipment does not permit enclosure within a water-vaporproof barrier as required in a method II package, packaging shall be accomplished by the alternate method specified in MIL-P-16298.

5.2.1.2 Packing. -

5.2.1.2.1 Levels A and B. -

5.2.1.2.1.1 Basic equipment. - The basic equipment shall be packed level A or B in accordance with MIL-P-16298.

5.2.1.3 Marking. - In addition to any special marking requirements, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.2.2 Repair parts, onboard. - Repair parts shall be preserved and packaged level A, packed level A, B, or C and marked in accordance with MIL-P-16298.

Source: https://assist.dla.mil -- Downloaded: 20260-19T19:43Z
Check the source to verify that this is the current version before use.

5.2.3 Manuals. - Manuals for electrical, electronic and mechanical equipment shall be prepared for shipment in accordance with MIL-M-15071.

5.2.4 Drawings. - Drawings shall be prepared for shipment in accordance with MIL-D-963.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Minimum controller output frequency (see 3.2.1.1(b), 3.2.1.11 and 3.5.4).
(c) Controller rating (see 3.2.1.1(c)).
(d) Dimensions and weight (see 3.3.5.2).
(e) Transducer electrical output (see 3.5.4).
(f) Transducer pressure input (see 3.5.5).
(g) Preservation, packaging, packing and marking requirements, if other than 5.1 (see 5.2).

Preparing activity:
Navy - SH
(Project 6110-N090SH)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

APPENDIX

INSPECTION CHECK LIST FOR IN-PROCESS INSPECTION

10. SCOPE

10.1 This appendix lists those items which should be included in the in-process inspection of the equipment furnished in accordance with this specification. This list is not an all inclusive list, and additions and deletions as applicable can be made by the manufacturer subject to the approval of the Government.

20. APPLICABLE DOCUMENTS

20.1 The following document of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 To facilitate the in-process inspection procedure, the manufacturer shall prepare a manufacturing flow chart showing the normal flow of material, parts, subassemblies, and assemblies through the manufacturing process. The manufacturer shall indicate on the flow chart the points at which the manufacturer conducts in-process inspection, and these points shall be designated as inspection stations.

30.1.1 For each inspection station the manufacturer shall prepare an inspection check sheet listing the items inspected at that station. Where appropriate, a standard sketch shall be included in the check sheet showing dimensions to be checked and the allowable tolerances, finish requirements, and so forth. Space shall be left on the sketch or on the check sheet to record the results of dimensional, visual or other inspection. A column shall be included on the check sheet for the initials or stamp of the manufacturer's representative conducting the inspection. A space shall be left on the inspection check sheet for the Government representative's initials or stamp. The inspection check sheet shall be suitably protected against damage or staining and will be physically attached to the item being inspected. When the component is assembled, the inspection check sheet for the component shall be attached to the assembly or otherwise made available. The words "approved drawing" as used herein mean a required drawing (class A drawing), which is approved by the cognizant approval activity, or a manufacturer's shop drawing if needed to depict details for in-process inspection not shown on a required drawing. Manufacturer's shop drawings which agree with approved required drawings may also be used for in-process inspection. Inspection requirements are not to be waived or reduced by selection of drawings.

30.2 General inspection items. - General inspection items shall be as follows:

(a) Material is as specified on approved drawing. Material was ordered in accordance with applicable material or component specification and was inspected in accordance with the requirements of the material or component specification.
(b) All welding is done by qualified welders and is in accordance with the approved drawing. There is no evidence of nonfusion, weld cracks, undersize welds, incomplete welds, heavy porosity, weld splatter or slag.
(c) All soldered connections are solidly bounded; there is no cold soldering, no rosin joints, no corrosive flux, no fractured joints or excess solder. Satisfactory connections were made prior to soldering. Bolted connections include approved locking devices and are secured against vibration. Solderless connectors are properly crimped, and the connector and crimping tool are proper size.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

(d) Finished castings are as shown on approved drawing and are clean and free of molding sand, cracks, blow holes, splits and deformations. Sufficient material is allowed for machining. Casting defects have not been covered by unauthorized repairs.
(e) Machining is as shown on approved drawing. The surfaces, including mating surfaces as applicable, are smooth, square and are free of burrs, sharp edges, tool marks, chatter marks, and scratches and damage due to handling. Surface finish is as shown on approved drawing, and there are no tool marks except those normally associated with the indicated surface finish. There are no flaws exposed as the result of machining.
(f) No parts are Government surplus or have been previously used or reclaimed.
(g) All items including hardware (nuts, bolts, lockwashers, and so forth) are made of corrosion resistant material or are given a corrosion resistant treatment.
(h) The Government may require a coil or winding to be cut apart to see the extent of varnish treatment and filling if there is any question as to the effectiveness of varnish treatment used.
(i) All bolts, nuts, set screws and other fasteners are secured in a manner which will preclude loosening in service. All locking devices shall meet the requirements of MIL-E-917.
(j) Insulation creepage and clearance distances are in accordance with approved drawing. Creepage distances are not achieved by means of cemented or butt joints.

30.4.1 Items to be inspected on major parts. -

(1) The following items were ordered in accordance with the applicable specification as modified by the contract requirements; are of the size, type and rating shown on the drawing; and are qualified products:

(a) Resistors and rheostats.
(b) Capacitors.

(2) The following items are not necessarily qualified products but are of the size, type and rating shown on the drawing:

(a) Switches.
(b) Rectifiers.
(c) Molded terminal boards.
(d) All laminated materials, such as glass mica glass and glass silicon glass used for terminal boards, sub-panels, and so forth.

(3) All materials used in the construction of reactors, magnetic amplifiers and transformers other than instrument transformers are as shown on the approved drawing and comply with the material requirements as shown on the approved drawing.
(4) All windings of reactors, magnetic amplifiers and transformers other than instrument transformers were given a minimum of three varnish treatments or one vacuum impregnation and bake followed by one dip and bake with an approved clear baking varnish in accordance with the procedures shown on the approved drawing.
(5) No black varnish, glyptal, paint or other treatment was given to the insulation of reactors, magnetic amplifiers, and transformers other than instrument transformers after the last dip and bake with clear baking varnish.
(6) The resistance of all windings of all reactors, magnetic amplifiers and transformers other than instrument transformers is as shown on the approved drawing. Ratio of turns is as specified on the approved drawing.
(7) The equipment enclosures are as shown on the approved drawing.

30.4.2 Items to be inspected on completed unit. -

(1) All components are mounted as shown on the approved drawing.
(2) All parts which require servicing, repair, replacement or periodic adjustment during the life of the equipment are readily accessible.
(3) All components are marked with identifying symbols, and these symbols agree with those used on the approved drawing.
(4) All terminals of all components and terminal boards are marked with identifying letters and numbers or both, and this identification agrees with those used on the approved drawing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

(5) Wire used for interconnecting components is of the type and size shown on the approved drawing.
(6) All wiring is neatly formed into groups and laced with nonflammable glass cord and supported or clamped in a manner which will prevent chafing of the insulation due to vibration and shock.
(7) Cable clamps used are of nonflammable material.
(8) There are no splices in any of the wires.
(9) Wire groups from hinged doors and panels are formed and clamped in a manner that sharp bends do not occur with the panel or door in either the open or closed position.
(10) Sufficient slack in wiring is allowed so that the weight of the harness is not supported by the terminal connections and so that at least two replacements can be made if lugs are clipped off at the wire end.
(11) Cable entrance provisions are as shown on the approved drawing.
(12) Sufficient space is provided to bring external cables through these cable entrances, support them within the enclosure and make connections at the appropriate terminals.
(13) Wherever wires run through holes in metal partitions or chassis, grommets are provided for mechanical protection.
(14) Wires are not bent around sharp corners which may injure the insulation.
(15) All wires are connected by either bolted or soldered connections.
(16) All bolted connections are provided with locking devices.
(17) The soldered connections do not rely on the solder for mechanical strength. The wire was mechanically secured before soldering and the solder was used only to get a good electrical path.
(18) No more than three wires are soldered to any one terminal.
(19) Both ends of all wires are marked with designations as shown on the approved drawing.
(20) Hinged doors and panels do not bind when opening.
(21) All identification and instruction plates are furnished as shown on the approved drawing.
(22) All information on identification and instruction plates is legible.
(23) The overall and mounting dimensions of the equipment are as shown on the approved drawing.
(24) The overall weight of the equipment is as shown on the approved drawing.
(25) The exterior and interior of the enclosure have been painted as shown on the approved drawing.
(26) All controls operate smoothly throughout their range.

Source: https://assist.dla.mil -- Downloaded: 20260-19T19:43Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the s
ification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for
taining information on the use of this specification which will insure that suitable products can be procured wi
minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fol
lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM, APR 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 20260-19T19:43Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
BUREAU OF SHIPS
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

CHIEF BUREAU OF SHIPS
SPECIFICATIONS AND STANDARDIZATION BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:43Z
Check the source to verify that this is the current version before use.